

2.5 Programming languages and Integrated Development Environments

Revise360

Teacher guide

Unit 2.5

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How to run this unit, lesson by lesson

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Start here

5 lessons. A realistic estimate is **5 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Language level lab p1-101	1	Lesson_Plans/PL_L01_LessonPlan.pdf
2	Language choice workshop p1-102	1	Lesson_Plans/PL_L02_LessonPlan.pdf
3	Translation studio p1-103	1	Lesson_Plans/PL_L03_LessonPlan.pdf
4	IDE control room p1-104	1	Lesson_Plans/PL_L04_LessonPlan.pdf
5	Language review HQ p1-105	1	Lesson_Plans/PL_L05_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard (`teacher.html`), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at `topics.html` with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- no optional challenge activity — every lesson
- every activity is recognition: the answer is always on the screen — p1-104

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Language level lab p1-101

Why might two programmers choose different languages to solve the same problem?

OUTCOMES	Compare high-level and low-level languages.; Explain why high-level source needs translation.; Justify language choices using project requirements.
STATIONS AND ACTIVITIES	6 stations, 14 activities, 27 marks (11 guided, 3 independent, 0 challenge)
RESOURCES	PL_L01_LanguageLevelLab_Lesson.pptx, PL_L01_LanguageLevelLab_Worksheet.docx
PLAN	Lesson_Plans/PL_L01_LessonPlan.pdf

2. Language choice workshop p1-102

Why might a school app and a small hardware routine need different language choices?

OUTCOMES	Compare development effort at different language levels.; Explain portability and hardware control.; Justify language choices using project requirements.
STATIONS AND ACTIVITIES	6 stations, 14 activities, 26 marks (12 guided, 2 independent, 0 challenge)
RESOURCES	PL_L02_InstructionWorkshop_Lesson.pptx, PL_L02_InstructionWorkshop_Worksheet.docx
PLAN	Lesson_Plans/PL_L02_LessonPlan.pdf

3. Translation studio p1-103

Why can a CPU not simply follow a Python source file as written?

OUTCOMES	Explain why programs need translation.; Compare how compilers and interpreters work.; Justify a translator choice using the task requirements.
STATIONS AND ACTIVITIES	6 stations, 14 activities, 24 marks (12 guided, 2 independent, 0 challenge)
RESOURCES	PL_L03_TranslationStudio_Lesson.pptx, PL_L03_TranslationStudio_Worksheet.docx
PLAN	Lesson_Plans/PL_L03_LessonPlan.pdf

4. IDE control room p1-104

What help would you want from the software you use to write programs?

OUTCOMES	Identify the four main IDE facilities.; Explain how each facility supports programming.; Use editing, diagnostics and execution tools in an IDE.
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STATIONS AND ACTIVITIES	6 stations, 14 activities, 23 marks (14 guided, 0 independent, 0 challenge)
RESOURCES	PL_L04_IDEControlRoom_Lesson.pptx, PL_L04_IDEControlRoom_Worksheet.docx
PLAN	Lesson_Plans/PL_L04_LessonPlan.pdf

5. Language review HQ p1-105

Which tool or language choice would most help you build a reliable revision app?

OUTCOMES Recall language levels and translator behaviour.; Select IDE tools for specific problems.; Write explanations linked to the needs of a scenario.

STATIONS AND ACTIVITIES	6 stations, 14 activities, 28 marks (12 guided, 2 independent, 0 challenge)
RESOURCES	PL_L05_LanguageReviewHQ_Lesson.pptx, PL_L05_LanguageReviewHQ_Worksheet.docx
PLAN	Lesson_Plans/PL_L05_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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